Storage Architecture

How Filex persists, replicates and serves files

Filex Demo Workspace

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Overview

This storage architecture document describes the layered design behind the Filex file service. Files are addressed by a stable content identifier, metadata lives in a relational store, and the bytes themselves are written to one or more pluggable backends.

The design goal is that any backend can be swapped without changing the API surface. The same request path handles a local disk on a laptop, an S3 bucket in production and a WebDAV mount in a customer data centre.

Backends

The primary production backend is S3-compatible object storage. Uploads are chunked into 8 MiB parts and committed with a multipart completion call, which lets interrupted uploads resume from the last confirmed part.

The WebDAV backend exists for customers with existing NAS infrastructure. It maps folder operations one-to-one onto PROPFIND, MKCOL and MOVE requests and caches directory listings for thirty seconds.

Local filesystem storage is used for development and for small single-node installations. Paths are canonicalised before every operation so that symlinks cannot escape the configured root.

Metadata

Every file has a row in the files table that records its size, checksum, MIME type and the backend key where the bytes live. Folder hierarchy is a materialised path column, which keeps listing a folder to a single indexed range scan.

Thumbnails and text extractions are stored as derived artefacts next to the original object under a derived/ prefix and are regenerated when the source checksum changes.

Replication and durability

Objects written to S3 rely on the provider's cross-zone replication. Local and WebDAV installations run a nightly rsync job that mirrors the storage root to a secondary volume and verifies checksums for a random one percent sample.

Deletes are soft for fourteen days. A background sweeper removes the underlying object once the retention window has passed and no share link still references the file.